THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 10, 22, 24, 6, 21, 2, 15 respectively, The mode of the data is: 1

(a) 242 **(b)** 264.4 **(c)** 240 **(d)** 260

2. The modal class is 20–30 with $f_1 = 16$, $f_0 = 6$, $f_2 = 8$. The mode is: 1

(a) $20 + \frac{8}{18} \times 10$ **(b)** $20 + \frac{10}{18} \times 10$ **(c)** $20 + \frac{10}{22} \times 10$ **(d)** $20 + \frac{10}{16} \times 10$

3. The mean of a distribution with classes 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180, 180–200 and frequencies 22, 12, 13, p, 5, 2, 8, 13 respectively is 106.4. The value of p is: 1

(a) 30 **(b)** 24 **(c)** 27 **(d)** 25

4. The median of a distribution with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 and frequencies 20, x, 16, y, 21, 5, 7 respectively, and total frequency 100, is 70. The value of x is: 1

(a) 22 **(b)** 23 **(c)** 11 **(d)** 21

5. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 19, 11, 8, 25, 8, 14, 15 respectively, The mean of the data is: 1

(a) 74.4 **(b)** 75 **(c)** 73.4 **(d)** 75.4

6. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 2, 22, 5, 20, 2, 16, 23, 10 respectively, The median of the data is: 1

(a) 65 **(b)** 63.8 **(c)** 70 **(d)** 60

7. For the data with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270, 270–280 with frequencies 6, 8, 18, 24, 15, 3, 6, 20 respectively, The mode of the data is: 1

(a) 241.7 **(b)** 230 **(c)** 234 **(d)** 240

8. If the mean of n classes' marks is found by the step-deviation method with $h = 10$ and $\sum f_i u_i = -12$, $\sum f_i = 60$, $a = 45$, the mean is: 1

(a) 45.2 **(b)** 33 **(c)** 47 **(d)** 43

9. **Assertion (A):** The modal class of a grouped frequency distribution is the class with the lowest frequency. 1
Reason (R): The mode is the value that occurs most often in the data.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The mean of 10, 20, 30, 40 and 50 is 30. 1
Reason (R): The sum of the deviations of the observations from their mean is zero.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	100–120	120–140	140–160	160–180	180–200	200–220	220–240	240–260
Number of workers	19	3	4	12	20	13	22	7

Find the mode.

12. The daily wages (in ₹) of workers in a factory are given below. **2**

Daily wage (₹)	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of workers	21	13	25	5	6	24	4	2

Find the median.

13. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 18, 13, 5, 16, p, 10, 15 respectively is 55.3. Find the value of p. **2**

14. The ages (in years) of patients admitted to a hospital in a month are given below. **2**

Age (years)	40–60	60–80	80–100	100–120	120–140	140–160	160–180	180–200
Number of patients	22	12	13	25	5	2	8	13

Find the mode.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The ages (in years) of patients admitted to a hospital in a month are given below. **3**

Age (years)	100–120	120–140	140–160	160–180	180–200	200–220	220–240	240–260
Number of patients	4	12	2	21	22	21	14	4

Find the mean and the mode (in this order).

16. The ages (in years) of patients admitted to a hospital in a month are given below. **3**

Age (years)	200–210	210–220	220–230	230–240	240–250	250–260	260–270	270–280
Number of patients	6	8	18	24	15	3	6	20

Find the mean.

17. The median of a distribution with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180 and frequencies 14, 17, x, 16, y, 13, 4, 12 respectively, and total frequency 100, is 93.75. Find the value of x. **3**

OR

The weights (in kg) of students of a class are given below. **3**

Weight (kg)	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Number of students	2	22	5	20	2	16	23	10

Find the mean and the mode (in this order).

18. The weights (in kg) of students of a class are given below. **3**

Weight (kg)	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of students	3	21	7	17	16	16	20

Find the mean.

OR

The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270, 270–280 and frequencies 21, 8, 8, 4, x, 24, y, 15 respectively, and total frequency 100, is 252.5. Find the value of x. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The monthly electricity consumption (in units) of 50 houses of a colony is: 100-200: 5 houses, 200-300: 10 houses, 300-400: 20 houses, 400-500: 8 houses, 500-600: 7 houses. **5**

(i) What is the value of $\frac{n}{2}$?
(ii) Find the cumulative frequency of each class.
(iii) Write the median class.
(iv) Find the median consumption.

20. The mean of a distribution with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 and frequencies 3, 13, p, 17, 15, 24, 17 respectively is 103.6. Find the value of p. **5**

OR

The marks obtained by students in a test are given below. **5**

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	3	4	23	4	15	10	19	22

Find the mean.

END OF QUESTION PAPER — Total: 40 Marks